

# Dev Patel

Senior Software Engineer • Cloud Platform • Event-Driven Systems • AI / Agent Platforms

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## SUMMARY

Senior-level cloud-platform and full-stack engineer with 5 years architecting cloud-native, event-driven systems for high-volume dental manufacturing. Owns systems end to end — AWS serverless, real-time IoT telemetry (1M+ daily events, sub-second visibility), and a production AI agent platform on Amazon Bedrock. Defines the reusable Terraform, serverless, authentication, and design-system patterns now standard across 40+ services and 13+ applications.

## EXPERIENCE

**Glidewell Dental** — *Software Engineer III* — Irvine, CA

**Sep 2021 – Present**

- **Own architecture through deployment** across a family of manufacturing platforms — MES, unified real-time data exchange (with HIPAA-oriented PII protection), fleet analytics, and authentication — setting patterns adopted org-wide.
- Established the standardized **Terraform + Lambda delivery pattern** — shared modules and GitLab CI that builds and deploys on apply, with multi-account promotion — now the default across **40+ backend services**, cutting new-service setup from days to hours.
- Built a production **AI agent platform on Amazon Bedrock** — a planner-routed, multi-model (Claude + Amazon Nova) tool-use agent with live DynamoDB access, a sandboxed code interpreter, and NL-to-SQL over operational data — letting non-technical staff query production data in natural language and cutting manual analyst and triage time.
- Architected **AWS IoT and Kafka/MSK pipelines processing 1M+ daily telemetry events**, cutting operational-visibility latency from minutes to sub-second and giving production facilities real-time fault visibility for the first time.
- Designed an **IoT device-management and OTA platform** for CNC milling machines: zero-touch X.509 provisioning, least-privilege per-device policies, over-the-air firmware/config delivery, and a device-shadow reconciliation engine with hash-verified config, conflict detection, and automated recovery.
- Delivered the serverless analytics behind a **3,500+ machine milling-fleet dashboard** — idle-machine detection, an Entity Hub 360 view, and Zendesk / Pega field-service integrations — unifying five facilities and nine production lines and eliminating hours of manual status checks daily.
- Built a **centralized authentication layer**: one reusable Lambda authorizer validating dual Cognito and Azure AD (MSAL) tokens plus API keys across HTTP, REST, and WebSocket APIs, standardizing secure access across internal applications.
- Led multiple **legacy-JavaScript** → **React 19 migrations** onto a shared MUI design system reused across 13+ dashboards.

## SELECTED SYSTEMS

**AI Agent Platform — Manufacturing Execution System.** Amazon Bedrock (Claude, Nova), Node.js, DynamoDB, API Gateway WebSockets. Planner-routed, multi-model agent with a 10-tool registry — DynamoDB access, a sandboxed code interpreter, analytics presets, and email/SNS/IoT actions — answering natural-language questions over live production data.

**Unified Real-Time Data Exchange Platform.** Python, AWS IoT Core, WebSocket API, DynamoDB, Timestream for InfluxDB, Terraform. Topic-routed pub/sub backbone ingesting all factory MQTT traffic and fanning out to webhooks, live WebSocket clients, and a time-series store, with a no-code message transformer and HIPAA-oriented PII hashing.

**3D Mesh QC Viewer.** Next.js, React, three.js. In-browser STL / PLY / OBJ rendering with reference-vs-test geometric deviation heatmaps and PDF/zip export for dental-part quality control.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript / Node.js, SQL; Java, C#, C / C++

**AWS:** Lambda, IoT Core, Greengrass, Kinesis, Kafka / MSK, DynamoDB, Timestream, S3, SQS / SNS / SES, API Gateway, Cognito, IAM, VPC, Amplify

**AI & ML:** Amazon Bedrock (Claude, Nova), agent orchestration & tool-use, NL-to-SQL over operational data, prompt engineering, production LLM integration

**Frontend:** React, MUI, Angular, Highcharts / Plotly, three.js, Vanilla-JS SPAs; Azure AD / MSAL

**Infra & DevOps:** Terraform, Kubernetes, GitLab CI/CD, Jenkins / CloudBees, Git; Azure, GCP; MS Teams, Zendesk, Pega, Upkeep, D365 / Dataverse

## EDUCATION

**Rutgers University** — B.S. in Computer Science, New Brunswick, NJ

**May 2021**